

Assignment 8 : Regression – Using Excel, and multiple Methods

This assignment is aimed at:

1. Learning to use Excel to create and evaluate regression models
 2. Get introduced to statistical tests used in residual analysis
 3. Getting introduced to and running various regression algorithms on a given data set and understanding their relative characteristics, performance, and advantages.
 4. Calculating, documenting, and understanding various regression metrics and developing an approach towards using them.
 5. Creating and consolidating multiple plots and tables with the aim of comparing and contrasting the results of the regression algorithms.
 6. Getting introduced to the relevant ML functions of the Python library: **sklearn**.
 7. Analysing the errors in ML models using residuals and QQ plots.
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Tasks:

- 1) Carry out the following tasks related to use of Excel for Linear Regression (Note: if you use any other spreadsheet, and not Excel, these tasks are still applicable - explore that spreadsheet's in-built statistical and regression functions)
 - a) Install and enable the **Data Analysis Toolpak**
 - b) Load **Assignment_8_newtons_law_time_distance_data_SLR.csv** into the spreadsheet and visualize the data using a scatter plot, and create **Descriptive Statistics** for the column 'y'
 - c) Create a SLR (Simple Linear Regression) model using the data in this data file. Create the following scatter plots, analyze them, and make conclusions:
 - i) y_{actual} and $y_{\text{predicted}}$ v/s x
 - ii) Error plot
 - d) Carry out the above regression steps using the following data sets, and critically analyze the outputs and errors and state your observations and conclusions:
 - i) **Assignment_8_newtons_law_time_distance_data_MLR.csv**
 - ii) **Assignment_8_newtons_law_time_distance_data_SLR2.csv**
- 2) Review the Notebook **Assignment_8_linear_regression_sklearn_OLS.ipynb**.
 - a) Focus on the plots created, understand and describe them.
 - b) Understand the different Python libraries and functions used to perform regression, and the outputs created by them.
 - c) In particular review the outputs created by the OLS function, and understand the **residual analysis** or **error analysis** related statistical test results.
- 3) Review the Jupyter Notebook **Assignment_8_multiple_regression_methods.ipynb**, along with the data file **Assignment_8_multiple_regression_methods.xlsx**:

- a) Execute the code. Analyze the resulting plots and metrics.
 - b) Choose the best algorithm for the given dataset. Justify your choice.
 - c) What is your take-away from the way the code has been structured to generate the results?
- 4) Review **sklearn** documentation for each sklearn function used in the Notebook (e.g., PolyNomialFeatures, LinearRegression, mean_squared_error, etc.) and create a summary of each to explain the functionality, the input parameters, and the outputs. Present this in the form of a two-column Table (Function name | Description).

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